Axiom's Stacks Project in Lean: what is verified, and what still needs review

On 4 November 2026, Axiom announced a Lean formalization of the Stacks Project, describing it as about 16 million lines of code with the reference's interconnected structure preserved. The announcement concerns a library of algebraic geometry, not a newly discovered theorem. Its most consequential claim is that later proofs can use that library without rebuilding the mathematics beneath them. [1]

Within two days, the release had prompted a detailed review by Lean developer and Mathlib maintainer Kim Morrison, public repair work, and a second Axiom release claiming formal proofs of the Weil conjectures on top of Stacks. Those developments make the initial announcement more interesting, but also harder to summarize as a single completed achievement. A checked formal proof, a faithful translation, and a maintained mathematical library are different accomplishments.

This account follows the public sources available on 7 November 2026, Europe/Rome time. The repositories and selected theorem statements have been inspected for this article, but their full Lean developments have not been independently rebuilt here.

A reference work becomes a proposed proof library

The original Stacks Project is a collaborative reference on algebraic geometry, maintained by Johan de Jong and its contributors. Its permanent four-character tags let a reader cite a particular definition or result without depending on a changing chapter number. Axiom's release builds on this human-written mathematics and on the Lean and Mathlib communities' existing formal infrastructure. Axiom credits de Jong with starting Stacks in 2005. [2]

Algebraic geometry studies geometric objects through algebraic equations. Modern proofs often depend on a long development of rings, schemes, sheaves, and cohomology. Translating an isolated theorem is of limited use if its definitions do not agree with the definitions used elsewhere. A shared Lean library could make those results composable, so that a later proof can import a theorem with the objects and hypotheses it actually needs.

Axiom says it used coordinated AI agents to translate statements, search Mathlib, construct proofs, and resolve dependencies. Its project metadata names Kenny Lau, Brian Nugent, Ashvin A. Swaminathan, and Yunzhou Xie as authors and maintainers. It describes a collaboration between human formalizers and AxiomProver, with foundational definitions and the reviewed library handled by people, and most of the unreviewed library formalized autonomously. That is a more precise attribution than saying that either humans or AI alone formalized Stacks. [3]

The release also has a narrower documented scope than the phrase "the entire Stacks Project" suggests. The metadata identifies a particular source commit and excludes exercises and obsolete chapters. The README leaves unformalized examples, exercises, and remarks as contributor tasks. Its tagging system explicitly permits a declaration to cover only part of a source tag. These qualifications matter when interpreting coverage. The 16-million-line figure is Axiom's reported size, not a measurement reproduced for this article. [4]

What Lean checks, and what review still has to establish

The repository separates `StacksCore`, which it describes as human-reviewed against the source and Mathlib conventions, from `StacksFrontier`, which contains AI-written formal-

izations awaiting that review. It reports kernel-checked proofs in both tiers, with no `sorry` placeholders and only Lean's standard axioms, `propext`, `Classical.choice`, and `Quot.sound`. The metadata labels the review status "self-assessed". Importing `Stacks` brings in both tiers. [4]

Lean checks whether a proof establishes its formal proposition under the definitions and assumptions in scope. It does not read the original prose and decide that the proposition is the intended theorem. Lean's own proof-validation guide makes that distinction explicit. It also explains why an axiom check and stronger independent checking can add assurance beyond an editor's successful proof indicator. [5]

A theorem can have no additional global axioms while still assuming the hard part as a local hypothesis. For example, a proof of "if this comparison map has a certain property, then this object is not algebraic" does not construct an object with that property. The implication may be correct and useful. It is not yet the promised example.

Axiom says its automated audits compare statements with the text and test their compatibility with downstream results. Such tests can expose mistakes. They are not an independent certification that every translation is faithful. The Core and Frontier distinction records an unfinished review task rather than two different kinds of proof logic.

A concrete gap in the non-algebraic Hom-stack example

In a 5 November review, Morrison welcomed the release while questioning its readiness as a reusable library. The review identifies duplicated Mathlib results, build warnings, and source statements whose formal counterparts do not establish the whole original claim. Morrison also notes that the gaps found were acknowledged in source comments. This is a criticism of coverage and usability, not an allegation that Lean accepted a false implication. [6]

The clearest example concerns Stacks tag 0AF8, a section constructing a Hom-stack that is not algebraic. A Hom-stack organizes maps between geometric objects, together with the isomorphisms between those maps. The example shows that favorable properties of the objects being mapped do not by themselves make this parameter space an algebraic stack. [7]

The setup is specific. Let k be an algebraically closed field that is not the algebraic closure of a finite field, and let S be the spectrum of the formal power-series ring $k[[t]]$. Take a projective, flat family X over S with geometrically connected curve fibres and regular total space. Its special fibre is nodal, has smooth irreducible components, and has a loop of rational curves in its dual graph. For a nonzero abelian variety A_0 over k, let A be the corresponding constant abelian scheme over S. Stacks proves that the Hom-stack of maps from X to the classifying stack $[S/A]$ is not algebraic.

The obstruction involves torsors, objects that locally resemble the group acting on itself but need not have a global choice of origin. The source constructs compatible torsors on infinitesimal thickenings of the special fibre that do not come from a torsor on the whole family. This failure of effectivity contradicts a necessary property of algebraic stacks in this setting.

The released Lean file does less. Its cohomological argument assumes the torsion, lifting, and non-torsion facts needed for the obstruction. Its final deduction assumes both the relevant algebraicity criterion and failure of essential surjectivity. The comments explicitly say that the Hom-stack itself is not defined and that the deduction is not instantiated at the source's example. The same file remained on the current branch when checked for this

article. [8]

This is a substantive missing part, not a cosmetic difference in notation. It also illustrates why counting tagged declarations cannot by itself establish complete coverage. The original Stacks section already discusses the example as a counterexample to a theorem of Aoki and its erratum. That counterexample is existing mathematics, not a new discovery by Axiom's agents.

Repairs are already part of the story

The repository history records a 5 November repair to `Germ.lean` and `DenseOpen.lean`, with a commit message saying it fixes a kernel rejection so that `lake build` succeeds. On 6 November, PR #7 merged an upgrade to Lean 4.35.0. The current `lean-toolchain` confirms that version. Morrison's criticism of the launch toolchain therefore should not be repeated as a description of the current branch. [9]

Morrison also opened concrete simplifications on 6 November. PR #9 replaces a recursive null-homotopy construction with Mathlib results, while PR #12 reuses an existing Hensel uniqueness theorem. Both were open at this article's cutoff. Their descriptions report local builds and axiom checks, but an open proposal is not an integrated repair. These changes address the practical cost of maintaining duplicate proofs without claiming that every coverage gap has been resolved.

The Weil conjectures provide a downstream test

On 6 November, Axiom announced a formalization of the Weil conjectures built on its Stacks library. The post credits Tony Yue Yu as lead contributor, with Letong Hong, Kenny Lau, Jannis Limperg, Brian Nugent, Ken Ono, Ashvin Swaminathan, Jesse Thorner, and Edison Xie. The repository says its roughly 980 modules were written by AI agents and have not yet been reviewed by humans. These credits and that review qualification belong together. [10]

The mathematics concerns counting solutions over finite fields. For a variety X over a field with q elements, write N_m for its number of points over the degree-m extension. Its zeta function packages all these counts into the formal series

$$Z(X,T) = \exp\left(\sum_{m\geq 1} N_m \frac{T^m}{m}\right).$$

For smooth projective varieties, the classical conclusions describe this series as a rational function with a functional equation, prescribe the magnitudes of the inverse roots of its factors, and relate their degrees to Betti numbers when there is a suitable complex variety with good reduction. These are established mathematical results. Formalizing them is not a new proof of the still-open classical Riemann hypothesis about prime numbers.

One particularly useful target in Axiom's repository uses only Mathlib definitions. Its hypotheses require a finite field and a smooth, proper, geometrically irreducible scheme of relative dimension d over that field. Smoothness excludes singularities. Properness is the algebraic condition analogous to compactness. Geometric irreducibility means irreducibility persists over an algebraic closure. Under these hypotheses, the statement expresses every finite-extension point count as an alternating sum of powers of algebraic integers, with absolute values $q^{i/2}$ in degree i, together with the expected duality. [11]

This target is also published by LeanEval. Having an external statement makes the intended conclusion easier to compare than a target expressed entirely through a new project's definitions. This particular statement does not contain the comparison with complex Betti numbers, which Axiom treats separately. A matching statement is still not the same evidence as an independently completed comparator run.

Axiom's README identifies the decisive weight estimate as `LAdicSheaf.weightBound`, Stacks tag 03VH, in `StacksFrontier`. Its account follows a later Fourier-transform route associated with Laumon rather than reproducing Deligne's 1974 proof line by line. The Weil manifest pins Stacks commit `012108f`, so the follow-up does not automatically inherit subsequent changes to Stacks main. This is concrete downstream use, but its assurance still depends on the unreviewed foundation it imports. [12]

A separate formal-arithmetic-geometry project is also public. Its authors are Bin Dong, Nailin Guan, Jiedong Jiang, Ruochuan Liu, Christian Merten, Xiangyu Pan, Tingxiao Shi, Bin Wu, and Wei Yang. Its metadata describes human-written headline statements and definitions, with most proof code written by supervised AI agents. It reports a second-team build from a shared cache and axiom checks, while saying that a full comparator run has not completed and that humans have not reviewed proof bodies line by line. These are that project's reported checks, not checks performed for this article. Its existence also makes an unsupported claim of priority inappropriate. [13]

The original Stacks Project remains separate

On 21 October, before Axiom's announcement, de Jong asked people not to submit AI-written comments or pull requests to the original Stacks Project. He welcomed contributions that people had noticed and typed themselves. That request is about the upstream project's contribution process. It should not be retold as a reaction to the 4 November release or as a blanket prohibition on separate formalization projects. Axiom's announcement acknowledges the boundary and says its formalization and proposed refinements stay in its own repository. [14]

Axiom also reports that its agents found dozens of source statements needing refinement. That aggregate remains a company claim in this account. Establishing individual corrections requires comparing the original statement, the proposed counterexample or missing hypothesis, and the repaired theorem. The existing Hom-stack counterexample should not be counted as newly found evidence for that claim.

The release has made substantial formal source code available for scrutiny and has already produced a downstream application and specific review proposals. Its broader promise is a reusable algebraic-geometry library. Assessing that promise now requires following which source statements are fully covered, which assumptions remain local hypotheses, and which reviews and repairs have actually landed. Neither the size of the code nor a successful check of one implication can answer those questions for the whole library.